

# Engine

Discrete-event simulation in Julia: machine\_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine\_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 | Generated: | Engine: DiscreteSim | Sections: 8

## The engine

A next-event calendar, coroutines that suspend on two channels, and statistics that know whether they are adequate.

### The run

Model machine\_shop  
Time unit minutes  
Seed 3690686459077356  
Events 7631  
Clock 3999  
Processes 618  
Statistics 13  
Streams 5  
Trace rows 6223  
Trace dropped 0  
Stop reason horizon

### Where the processes ended up

READY  
0

RUNNING  
0

WAITING  
4

BLOCKED  
77

PASSIVE  
0

FINISHED  
537

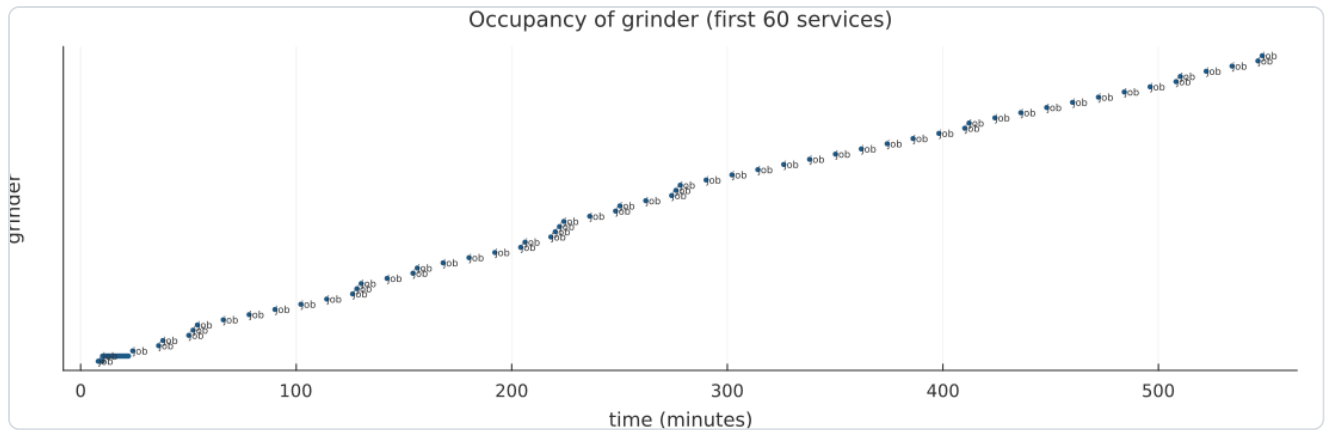
FAILED  
0

CANCELLED  
0

### Little's law, checked on every resource

Little law

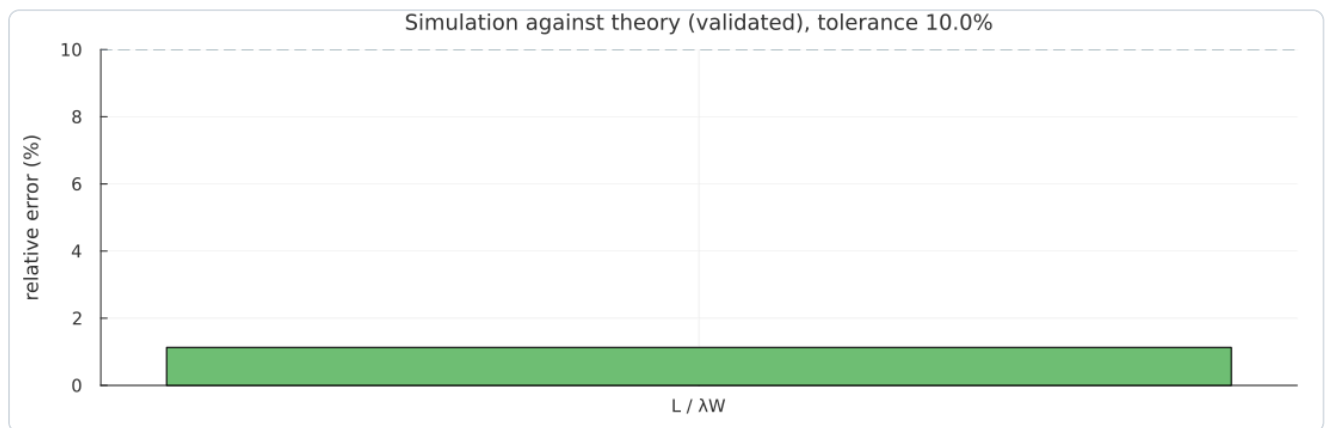
| Resource | L     | $\Lambda$ | W     | $\Lambda W$ | Relative error | Holds |
|----------|-------|-----------|-------|-------------|----------------|-------|
| mill     | 6.578 | 0.092     | 75.6  | 6.971       | 0.06           | no    |
| drill    | 34.6  | 0.122     | 286   | 35          | 0.011          | yes   |
| grinder  | 36.1  | 0.103     | 325.4 | 33.5        | 0.07           | no    |
| lathe    | 1.535 | 0.083     | 18.5  | 1.537       | 0.00121        | yes   |
| press    | 1.069 | 0.079     | 13.4  | 1.066       | 0.00208        | yes   |



Occupancy

## Against theory

Verdict: **validated** worst relative error 1.13% against a tolerance of 10%.



Validation

Cross checks

| Observed key | Theory key  | Observed | Theory | Relative error | Within tolerance |
|--------------|-------------|----------|--------|----------------|------------------|
| L            | $\lambda W$ | 34.6     | 35     | 0.011          | yes              |